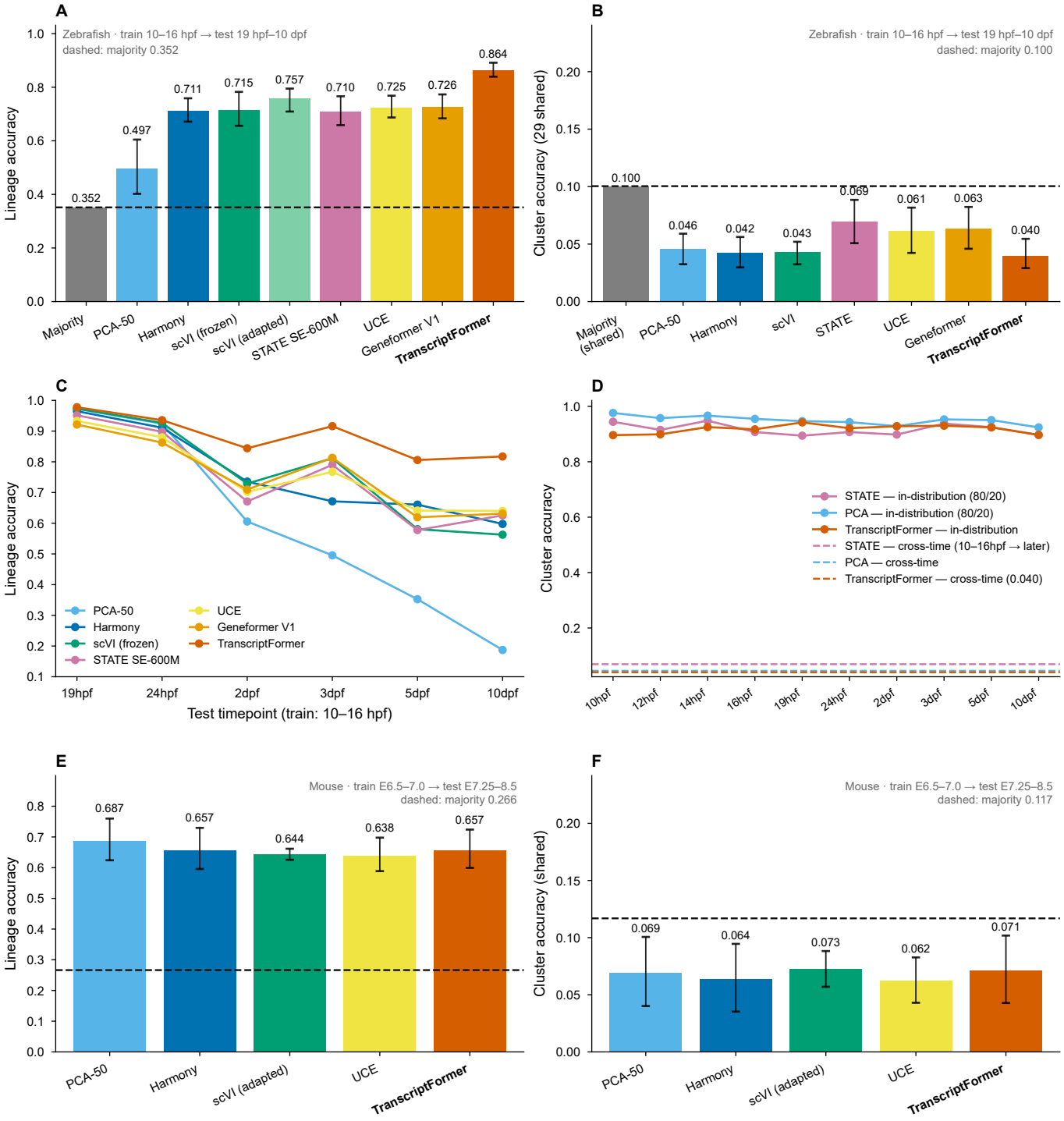

A
Zebrafish · train 10–16 hpf → test 19 hpf–10 dpf
dashed: majority 0.352
Lineage accuracy
0.352
0.497
0.711
0.715
0.757
0.710
0.725
0.726
0.864
Majority
PCA-50
Harmony
scVI (frozen)
scVI (adapted)
STATE SE-600M
UCE
Geneformer V1
TranscriptFormer
B
Zebrafish · train 10–16 hpf → test 19 hpf–10 dpf
dashed: majority 0.100
Cluster accuracy (29 shared)
0.100
0.046
0.042
0.043
0.069
0.061
0.063
0.040
Majority (shared)
PCA-50
Harmony
scVI
STATE
UCE
Geneformer
TranscriptFormer
C
Lineage accuracy
PCA-50
Harmony
scVI (frozen)
STATE SE-600M
UCE
Geneformer V1
TranscriptFormer
19hpf
24hpf
2dpf
3dpf
5dpf
10dpf
Test timepoint (train: 10–16 hpf)
D
Cluster accuracy
STATE — in-distribution (80/20)
PCA — in-distribution (80/20)
TranscriptFormer — in-distribution
STATE — cross-time (10–16hpf → later)
PCA — cross-time
TranscriptFormer — cross-time (0.040)
10hpf
12hpf
14hpf
16hpf
19hpf
24hpf
2dpf
3dpf
5dpf
10dpf
E
Mouse · train E6.5–7.0 → test E7.25–8.5
dashed: majority 0.266
Lineage accuracy
0.687
0.657
0.644
0.638
0.657
PCA-50
Harmony
scVI (adapted)
UCE
TranscriptFormer
F
Mouse · train E6.5–7.0 → test E7.25–8.5
dashed: majority 0.117
Cluster accuracy (shared)
0.069
0.064
0.073
0.062
0.071
PCA-50
Harmony
scVI (adapted)
UCE
TranscriptFormer